

Retrieval token accuracy by context length and needle depth

NoPE — Untouched (0.6% @256K)

	2K	4K	8K	16K	32K	64K	128K	256K
10%	100	94	49	48	34	12	4	0
50%	99	84	49	48	34	14	6	1
90%	99	50	49	47	37	15	6	1

Polar — Untouched (34.4% @256K)

	2K	4K	8K	16K	32K	64K	128K	256K
10%	99	79	67	52	66	57	41	30
50%	99	99	98	83	69	50	44	34
90%	98	98	97	84	68	56	40	39

RoPE — Untouched (0.0% @256K)

	2K	4K	8K	16K	32K	64K	128K	256K
10%	81	37	23	1	10	0	0	0
50%	88	7	18	0	0	0	0	0
90%	96	30	11	0	0	0	0	0

NoPE — One-head cap (0.9% @256K)

	2K	4K	8K	16K	32K	64K	128K	256K
10%	99	70	55	45	33	14	7	1
50%	99	99	92	47	35	17	10	1
90%	98	98	97	49	39	18	10	1

Polar — One-head cap (47.1% @256K)

	2K	4K	8K	16K	32K	64K	128K	256K
10%	99	98	98	81	82	68	58	41
50%	99	98	97	77	82	62	67	50
90%	98	98	96	74	86	77	66	50

RoPE — One-head cap (0.0% @256K)

	2K	4K	8K	16K	32K	64K	128K	256K
10%	56	33	21	0	6	0	0	0
50%	78	4	13	0	0	0	0	0
90%	95	27	4	0	0	0	0	0

Raven Native — Reference (20.3% @256K)

	2K	4K	8K	16K	32K	64K	128K	256K
10%	31	28	28	24	22	20	20	19
50%	36	37	28	24	23	21	20	19
90%	56	45	36	29	29	24	24	23

Atma-Raven-Titans — Reference (10.0% @256K)

	2K	4K	8K	16K	32K	64K	128K	256K
10%	30	30	26	22	18	13	8	4
50%	32	29	25	21	19	16	12	8
90%	67	36	30	28	27	25	21	18